# Observationally Calibrated Sea Level Rise Projections to 2150

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**Abstract**

We present global mean sea level (GMSL) projections to 2150 derived from an observationally calibrated rate–temperature relationship. Dynamic ordinary least squares (DOLS) applied to seven GMSL records and four global mean surface temperature (GMST) products yields a robust quadratic relationship between the rate of sea level rise and warming: the thermodynamic transient sensitivity $\mathrm{d}\alpha/\mathrm{d}T = 2.85 \pm 0.38$ mm/yr/°C$^2$ across the thermodynamic dataset ensemble. Projections under SSP1-2.6 through SSP5-8.5, produced by evaluating this relationship at FaIR-emulated temperature trajectories, exceed IPCC AR6 process-model-based thermodynamic projections by [X]% under SSP2-4.5 and [Y]% under SSP5-8.5 at 2100. The difference arises because CMIP-class ocean models produce a genuinely more linear rate–temperature relationship, with transient sensitivity approximately half the observational value. The observationally calibrated projection decomposes naturally into a thermodynamic component (constrained by the DOLS calibration) and an ice-sheet residual, providing an independent bound on cryospheric contributions.

# 1 Introduction

# 2 Observational Calibration

## 2.1 Dynamic OLS rate–temperature model

## 2.2 Multi-dataset robustness

## 2.3 Epoch dependence

# 3 Projections

## 3.1 Method

## 3.2 DOLS projections

## 3.3 Decomposition into thermodynamic and residual

# 4 Comparison with IPCC AR6 / FACTS

# 5 Discussion

## 5.1 Strengths and limitations

## 5.2 Implications for AR7

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